SECP3213-01

BUSINESS INTELLIGENCE

PHASE 3

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Impact of Severe Weather Events on US Commercial Flight Operations: A 2016 Big Data Analytics Approach

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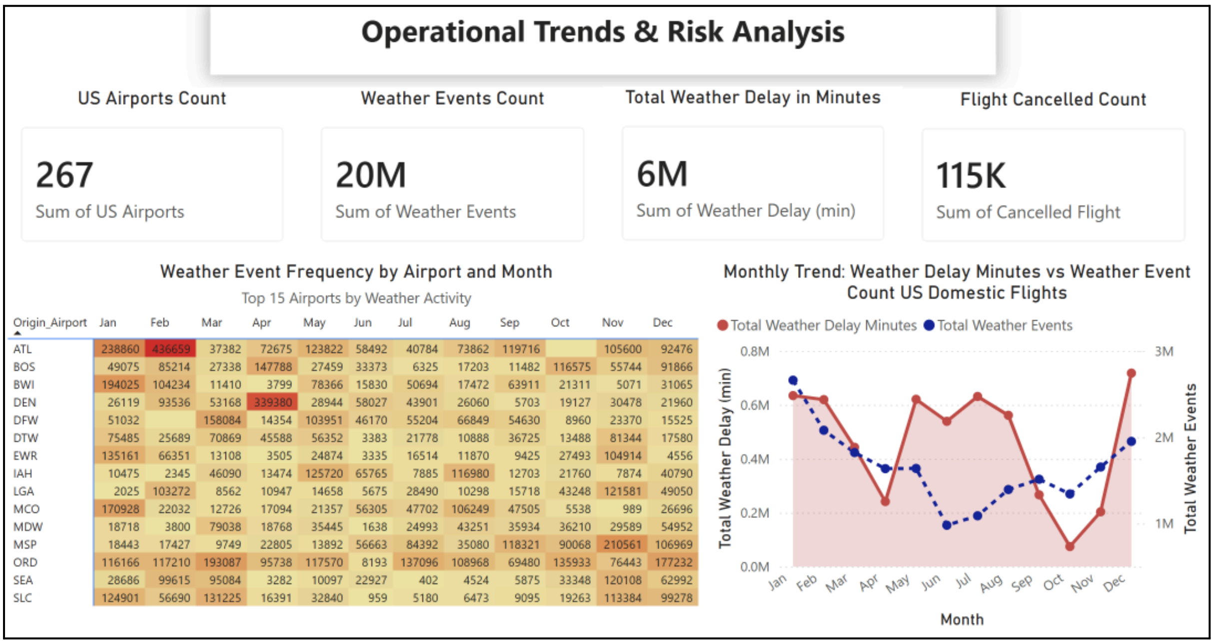
1.0 Visualization & Dashboard

1.1 Dashboard 1 : Operational Trends & Risk Analysis

The dashboard provides a temporal analysis of the distribution of weather disruption events in the US air transportation system in 2016. The four KPIs include: US Airports Count, Weather Events Count, Total Weather Delay Minutes, and Flight Cancelled Count. Heat map is used to show the frequency of weather disruptions per airport and month to allow quick identification of the high risk airport-month combinations. Additionally, there is a trend line that plots the number of monthly weather events against total weather delay minutes in order to allow for seasonality identification and correlation between weather frequency and impact on operations.



Origin_Airport	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
ATL	238860	436659	37382	72675	123822	58492	40784	73862	119716		105600	92476
BOS	49075	85214	27338	147788	27459	33373	6325	17203	11482	116575	55744	91866
BWI	194025	104234	11410	3799	78366	15830	50694	17472	63911	21311	5071	31065
DEN	26119	93536	53168	339380	28944	58027	43901	26060	5703	19127	30478	21960
DFW	51032		158084	14354	103951	46170	55204	66849	54630	8960	23370	15525
DTW	75485	25689	70869	45588	56352	3383	21778	10888	36725	13488	81344	17580
EWR	135161	66351	13108	3505	24874	3335	16514	11870	9425	27493	104914	4556
IAH	10475	2345	46090	13474	125720	65765	7885	116980	12703	21760	7874	40790
LGA	2025	103272	8562	10947	14658	5675	28490	10298	15718	43248	121581	49050
MCO	170928	22032	12726	17094	21357	56305	47702	106249	47505	5538	989	26696
MDW	18718	3800	79038	18768	35445	1638	24993	43251	35934	36210	29589	54952
MSP	18443	17427	9749	22805	13892	56663	84392	35080	118321	90068	210561	106969
ORD	116166	117210	193087	95738	117570	8193	137096	108968	69480	135933	76443	177232
SEA	28686	99615	95084	3282	10097	22927	402	4524	5875	33348	120108	62992
SLC	124901	56690	131225	16391	32840	959	5180	6473	9095	19263	113384	99278

Figure 1.1 Dashboard 1 : Operational Trends & Risk Analysis

The dashboard reveals that weather-related disruptions are concentrated at major airports such as ATL, ORD, BWI, and DEN, particularly during winter months. The frequency of weather incidents and delay times reach their peaks in January and again towards the end of the year. This is an indication of the existence of seasonal effects on the weather impacts at the airports. Airlines and airport management teams are advised to undertake resource planning for specific seasons by ensuring availability of resources in the affected airports during the winter period.

1.2 Dashboard 2 : Geographic & KPI Overview

The dashboard shows the geographical breakdown of the impacts of weather disruption and how they affect airport performance. It uses a bubble map to visualize the total number of weather events for each airport within the US, and this makes it easy to see where regions with high weather activity can be found. The clustered bar chart helps to compare the average departure delay for each weather disruption risk level in the top ten busiest airports.

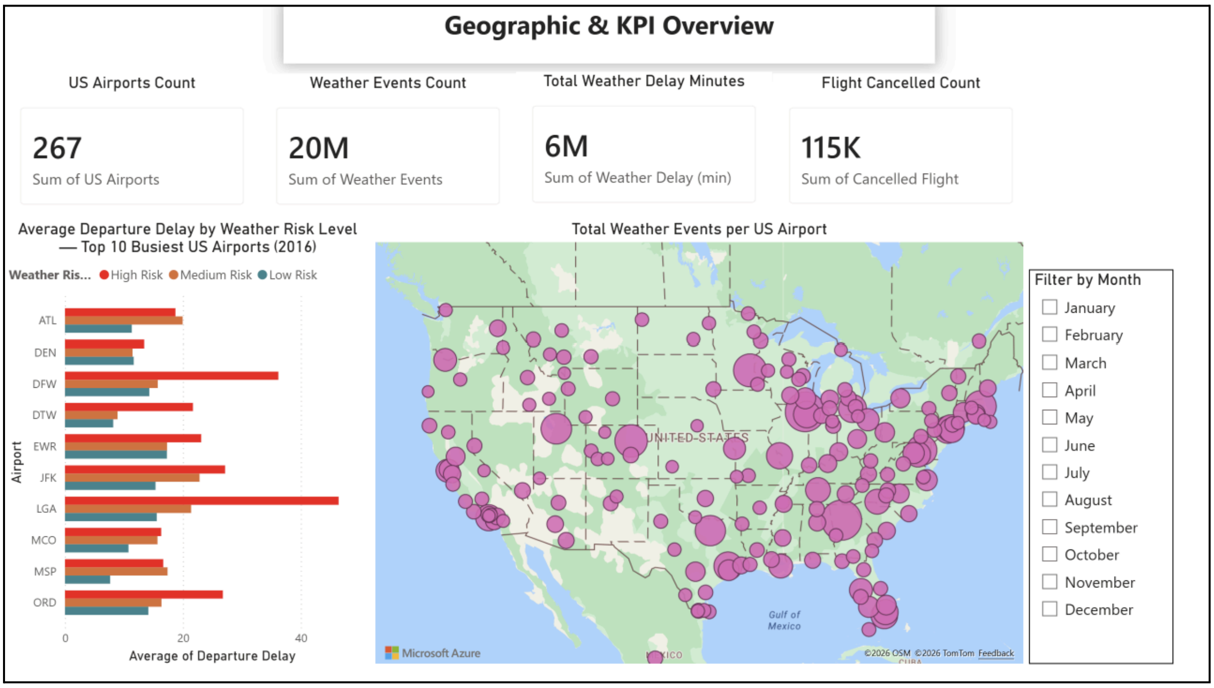


Figure 1.2 Dashboard 2 : Geographic & KPI Overview

From the analysis above, it is evident that weather disturbances occur in areas like the Midwest, Texas, and the Northeast. Nonetheless, it is important to note that airports experiencing frequent weather disruptions may not be those that experience maximum delays. The example here is LGA and DFW, which record maximum sensitivity to weather disruptions even though they experience few cases compared to ATL and ORD. It is thus critical for airport administrators to focus on airports with maximum sensitivity to weather disturbance to enhance their operations and minimize delays.

1.3 Storyboard: Weather vs Flight Delays

The storyboard creates an analytical narrative which describes in structured terms the impact of weather on flight delays. Visualisation is carried out by four graphs which are sequentially placed to address key analytical questions. These include a dual axis line graph for comparing weather related delays to overall delays, a donut chart to show the percentage breakdown of weather types, a stacked bar graph to indicate which airports have high weather delays and a scatter plot to investigate the correlation between the amount of rain and time taken to delay flights. This storytelling approach guides users through the analysis process and provides a clearer understanding of weather-related operational challenges.

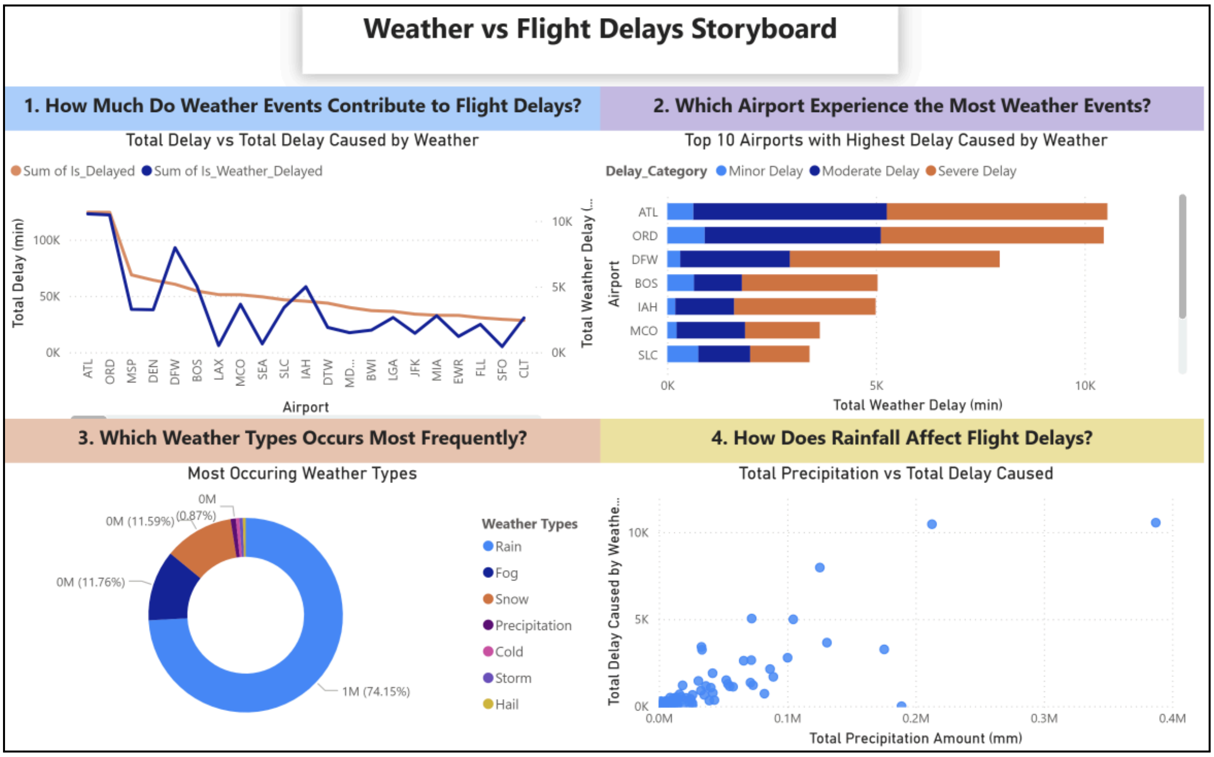


Figure 1.3 shows Storyboard : Weather vs Flight Delays

The results from the storyboard suggest that the weather is a major cause of flight delays, although there are several other causes too. The most prominent cause among these is rain, followed by snow and fog. ATL and ORD are examples of airports that have been affected the most due to weather, and thus require more focus in managing potential disruptions. The analysis of the rain data shows that heavy rainfall results in larger delays. It is therefore recommended that airlines and airport operators should adopt threshold-based weather response strategies and strengthen forecasting capabilities to improve operational resilience and reduce disruption impacts.